

Linear Algebra Problem Set 13

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Exercise 5.15

Since A is orthogonal, assume λ is an eigenvalue of A , then

$$Ax = \lambda x$$

for some x .

Then

$$\|Ax\| = \|\lambda x\| = \|x\|$$

by definition of orthogonal matrix.

Then

$$\|\lambda x\| = |\lambda| \|x\| = \|x\|$$

implies

$$|\lambda| = 1 \Rightarrow \lambda \in \{-1, 1\}.$$

So the only possible eigenvalues are 1 and -1 .

Since A is diagonalizable, then there exists a matrix P such that P is invertible, and

$$A = PDP^{-1}.$$

As we know that $\det(A) = 1$, hence

$$\begin{aligned}\det(A) &= \det(PDP^{-1}) \\ &= \det(P) \det(D) \det(P^{-1}) \\ &= \det(P) \det(D) \frac{1}{\det(P)} = \det(D).\end{aligned}$$

And since

$$D = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

where λ_1, λ_2 are eigenvalues of A ,

$$\det(A) = \det(D) = \lambda_1 \lambda_2 = 1.$$

Since $\lambda_1, \lambda_2 \in \{-1, 1\}$ and $\lambda_1 \lambda_2 = 1$, then

$$D = I \quad \text{or} \quad D = -I.$$

If $D = I$, then

$$A = PIP^{-1} = PP^{-1} = I.$$

If $D = -I$, then

$$A = P(-I)P^{-1} = -PP^{-1} = -I.$$

Exercise 5.6

Since L is an isomorphism, L^{-1} exists.

Then consider

$$M \circ L = L^{-1} \circ (L \circ M) \circ L.$$

Then, let's compute the characteristic polynomial of $M \circ L$.

$$\begin{aligned}\det(M \circ L - \lambda I) &= \det(L^{-1} \circ (L \circ M) \circ L - \lambda I) \\ &= \det(L^{-1} \circ (L \circ M - \lambda I) \circ L) \\ &= \det(L^{-1}) \det(L \circ M - \lambda I) \det(L).\end{aligned}$$

Since L is invertible,

$$= \frac{1}{\det(L)} \det(L \circ M - \lambda I) \det(L) = \det(L \circ M - \lambda I).$$

Hence

$$\det(L \circ M - \lambda I) = \det(M \circ L - \lambda I).$$

Exercise 5.8

Let

$$A = \begin{pmatrix} p & -q \\ q & p \end{pmatrix}, \quad p, q \in \mathbb{R}, \quad q \neq 0.$$

$$\begin{aligned}\det(A - \lambda I) &= \det \begin{pmatrix} p - \lambda & -q \\ q & p - \lambda \end{pmatrix} \\ &= (p - \lambda)^2 + q^2 > 0\end{aligned}$$

over $\mathbb{R}$, since $q \neq 0$.

So there are no real eigenvalues over $\mathbb{R}$.

Over $\mathbb{C}$,

$$\begin{aligned}(p - \lambda)^2 + q^2 &= (p - \lambda - qi)(p - \lambda + qi) \\ &= ((p - qi) - \lambda)((p + qi) - \lambda).\end{aligned}$$

Hence

$$\lambda_1 = p - qi, \quad \lambda_2 = p + qi.$$

Then A has two distinct eigenvalues, so it is diagonalizable over $\mathbb{C}$.

For $\lambda_1 = p - qi$,

$$Ax = (p - qi)x$$

means

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (p - qi) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

So

$$\begin{pmatrix} px_1 - qx_2 \\ qx_1 + px_2 \end{pmatrix} = \begin{pmatrix} (p - qi)x_1 \\ (p - qi)x_2 \end{pmatrix}.$$

Thus

$$qx_2 = qix_1 \Rightarrow x_2 = ix_1,$$

and

$$qx_1 = -qix_2 \Rightarrow x_1 = -ix_2.$$

Therefore

$$V_{\lambda_1}^A = \text{span} \left\{ \begin{pmatrix} 1 \\ i \end{pmatrix} \right\}.$$

For $\lambda_2 = p + qi$,

$$Ax = (p + qi)x$$

gives us

$$x_2 = -ix_1, \quad x_1 = ix_2.$$

Therefore

$$V_{\lambda_2}^A = \text{span} \left\{ \begin{pmatrix} 1 \\ -i \end{pmatrix} \right\}.$$

Exercise 5.19

(i)

Let $\lambda \in \mathbb{C}$ be an eigenvalue of L . This means

$$L(x_0, x_1, x_2, \dots) = \lambda(x_0, x_1, x_2, \dots),$$

which means

$$(x_1, x_2, \dots) = (\lambda x_0, \lambda x_1, \dots).$$

Therefore

$$x_{i+1} = \lambda x_i$$

for all $i = 0, \dots, \infty$.

And this is exactly the definition of a complex infinite geometric sequence.

And surely, for every $\lambda \in \mathbb{C}$, there exists a geometric sequence such that

$$x_{i+1} = \lambda x_i.$$

Hence every $\lambda \in \mathbb{C}$ is an eigenvalue of L .

The eigenspace V_λ^L associated with λ is all sequences $(x_n)_{n=0}^\infty$ such that

$$x_{i+1} = \lambda x_i$$

for all $i = 0, \dots, \infty$.

If $x_0 \neq 0$, then

$$x_n = \lambda^n x_0.$$

So

$$V_\lambda^L = \text{span}\{(1, \lambda, \lambda^2, \dots)\}.$$

(ii)

$$R(x_0, x_1, \dots) = (0, x_0, x_1, \dots).$$

If λ is an eigenvalue of R , then

$$\begin{aligned} R(x_0, x_1, \dots) &= \lambda(x_0, x_1, \dots) \\ &= (0, x_0, x_1, \dots). \end{aligned}$$

Entrywise,

$$\lambda x_0 = 0,$$

$$\lambda x_1 = x_0,$$

$$\lambda x_2 = x_1.$$

So

$$\lambda x_i = x_{i-1}$$

for all $i = 1, \dots, \infty$.

Let $x_{-1} = 0$.

Then let's consider two cases.

Case 1: $\lambda = 0$. Then

$$x = (0, \dots, 0),$$

which contradicts that λ is an eigenvalue of R .

Case 2: $\lambda \neq 0$.

Then

$$\lambda x_0 = 0 \Rightarrow x_0 = 0.$$

Consequently,

$$\lambda x_1 = 0 \Rightarrow x_1 = 0,$$

and generally

$$\lambda x_i = 0$$

for all $i = 0, \dots, \infty$.

Hence

$$x_i = 0$$

for all $i = 0, \dots, \infty$, which contradicts that x is an eigenvector.

Therefore, by two cases,

$$\sigma(R) = \emptyset.$$